Module 1 - 3

Assignment - 1

1. Define Data science and explain it's life cycle
2. Differentiate descriptive, Diagnostic, predictive & Prescriptive Analytics
3. Explain the steps involved in Data preprocessing with Examples

Answers

1. Definition:

Data Science is study of collecting, processing and analyzing data to gain aseful information & make better decisions.

life cycle:

1. problem Definition - identify the problem
2. Data collection - collect data from different Sources
3. Data Analysis - Analyze the data to find patterns
4. Data preprocessing - clean and organize the data
5. Model Building - Build a machine learning
6. Evaluation & Deployment - Test the model & use it in real applications.

2.

Analytics Type	purpose	Question Answered	Example
Desariptive Analytics	Summarize past data	What happened	Monthly sales report
Diagnostic Analytics	finds the reasons behind events	Why did it happen?	Sales dropped because demand decreased

Predictive Analytics	Predicts future outcomes	What will happen?	forecast next Month's Sales
Prescriptive Analytics	Suggests the best action	What should be done?	Recommed discount offers to increase Sales.

5. Definition:

Data preprocessing is the process of cleaning and preparing raw data before analysis

Steps:

1. Data cleaning - Remove Missing values and errors
   Ex: fill missing age with the average age
2. Data integration - Combine data from different sources
3. Data transformation - convert data into the required format
   Ex: Male = 1, Female = 0
4. Data Reduction - Remove unnecessary data
5. Data validation - check wheather the data is correct and Consistent.